

Claire Mundi

mundi@wisc.edu

clairemundi.github.io

PhD Candidate

University of Wisconsin-Madison

Department of Atmospheric & Oceanic Sciences

Education

- 2022–2026* **Ph.D.**, Atmospheric & Oceanic Sciences, University of Wisconsin, Madison, WI
2020–2022 **M.S.**, Atmospheric & Oceanic Sciences, University of Wisconsin, Madison, WI
2016–2020 **B.A.** Physics, Geosciences, Skidmore College, Saratoga Springs, NY
Summa Cum Laude | *Phi Beta Kappa*, *Sigma Pi Sigma*, *Pi Mu Epsilon* Honor Societies

Research Experience

- 2020– **Graduate Research Assistant**
[The Atmospheric Radiation and Climate Research Group](#)
University of Wisconsin-Madison (Advisor: Dr. Tristan L'Ecuyer)
Analyzed the role of intense cyclones on the marginal ice zone in both hemispheres using a combination of satellite observations, reanalyses, and climate model output.
- 2019–2020 **Undergraduate Research Assistant**
Skidmore College (Advisor: Dr. Greg Gerbi)
Investigated particle transport pathways for two species of sea snail larvae in the Delaware Bay Region using a coupled hydrodynamic model (ROMS) and a Lagrangian particle-tracking model (LTRANS).
- 2018 **Scripps Undergraduate Research Fellowship (SURF)**
Scripps Institution of Oceanography (Advisor: Dr. Dan Lubin)
Determined 3 years of cloud phase properties over the Greenland Ice Sheet using ground-based micropulse lidar signals from Summit Station.

Awards & Honors

- 2020 Eleanor A. Samworth Prize for Excellence in Physics (Skidmore College)
2016–20 Porter Presidential Scholar in Science and Mathematics (Skidmore College)

Publications

 [Google Scholar](#)

*Expected.

Journal Articles

- J1. **Mundi, Claire** & L'Ecuyer, T. Is the Modern Arctic Marginal Ice Zone More Susceptible to Summer Cyclones? *Journal of Climate* **38**, 403–422 (2025).
- J2. **Mundi, Claire**, L'Ecuyer, T. & DuVivier, A. Have Impacts of Intense Arctic Cyclones on Summer Sea Ice Reached a Maximum? *Geophysical Research Letters* **52**, e2025GL117848 (2025).

Working papers

- W1. **Mundi, Claire.** & L'Ecuyer, T. *Intense Cyclones Increase Ice Area Within the Global Marginal Ice Zone*. In prep. 2026.

Presentations

Talks

- T1. **Mundi, Claire** & L'Ecuyer, T. *An Observed Seasonal Symmetry in Intense Cyclone Impacts on the Global Marginal Ice Zone* AGU Annual Meeting (New Orleans, LA, USA & Online). Dec. 2025.
- T2. **Mundi, Claire** & L'Ecuyer, T. *Observed and Predicted Changes in Arctic Sea Ice Response to Intense Summer Cyclones* AMS 18th Conference on Polar Meteorology and Oceanography (Denver, CO, USA). May 2025.
- T3. **Mundi, Claire** & L'Ecuyer, T. *How Do Intense Summer Cyclones Impact the Arctic Marginal Ice Zone in the CESM2 Large Ensemble?* NCAR Polar Climate Working Group Meeting (Denver, CO, USA). Mar. 2025.

Posters

- P1. **Mundi, Claire** & L'Ecuyer, T. *An Observed Transition to More Marginal Sea Ice Loss Due to Arctic Summer Cyclones* AGU Annual Meeting (Washington, D.C., USA). Dec. 2024.
- P2. **Mundi, Claire** & L'Ecuyer, T. *Role of Cyclone-Induced Near-Surface Winds and Sea Surface Temperature Changes on Arctic Sea Ice* AMS 17th Conference on Polar Meteorology and Oceanography (Madison, WI, USA). Aug. 2022.
- P3. Beckley, I., **Mundi, Claire**, L'Ecuyer, T. & Martin, J. *Analysis of the Kinematic and Radiative Impacts on Sea Ice of a Robust Late-Summer Cyclogenesis Event in the Arctic*. AMS Annual Meeting, Conference on Mesoscale Processes (Remote). Jan. 2022.
- P4. **Mundi, Claire**, Wilson, J., Gerbi, G., Fuchs, H., Hunter, E., Chant, R. & Wilkins, J. *Particle pathways in the mid-Atlantic bight*. AGU Ocean Sciences Meeting (San Diego, CA, USA). Feb. 2020.
- P5. **Mundi, Claire** & Lubin, D. *Frequency of Radiatively Active Liquid Water Clouds Above the Greenland Ice Sheet*. Summer Symposium, Scripps Institution of Oceanography (La Jolla, CA, USA). Aug. 2018.

Outreach

- O1. **Mundi, Claire.** *Observed and Predicted Changes in Arctic Sea Ice Response to Intense Summer Cyclones: Research Overview* FIRE Climate Computing, University of Maryland (Remote). June 2025.
- O2. **Mundi, Claire.** *Physics Alumni Research Talk.* Skidmore College (Remote). Feb. 2023.

Teaching

Skidmore College

- 2017–2020 *Peer Academic Coach*
General Physics I, General Physics II, Modern Physics, Thermal and Statistical Physics, Calculus II, Atmosphere and Ocean Dynamics, Mathematical and Computational Methods in Physics
- 2019–2020 *Peer Mentor for First-Year Seminar*
The Deep Seafloor: Facilitated discussions on ocean observations and exploration.

Service and Outreach

- 2020–2025 **AOS Mentorship Program**, University of Wisconsin, Madison, WI
Mentored undergraduate students through monthly meetings.
Led department-wide networking and writing workshops.
- 2020–2025 **AOS Graduate Student Association**, University of Wisconsin, Madison, WI
Performed public science demonstrations to over 1,000 students across several events including the WI Science Festival and UW Science Expeditions.
Alumni Engagement Board Representative (2021–2022)
- 2022–2024 **STORM REU**, University of Wisconsin, Madison, WI
Organized professional development events for three cohorts of visiting undergraduate summer research students.
- 2022– *Letters to a Pre-Scientist* Volunteer
Served as a STEM pen-pal to three middle school students from low-income areas, addressing topics such as education pathways, overcoming barriers, and various career opportunities in STEM.
- 2023– *Adopt-a-Physicist* Volunteer
Responded to over 40 questions per year from high school physics students about Arctic climate, graduate programs, and careers in science.

2025– **Journal Article Reviewer**
Journal of Geophysical Research - Atmospheres

Trainings and Workshops

2025 **Scalable and Computationally Reproducible Approaches to Arctic Research**,
National Center for Ecological Analysis and Synthesis, Santa Barbara, CA

2024 **IARPC Collaborations Mentorship Program**

2024 **NOAA Women in Sciences Leadership Workshop**, Geophysical Fluid Dynamics
Laboratory, Princeton, NJ

2023 **CESM Tutorial**, National Center for Atmospheric Research, Boulder, CO

Last updated: December 2025